

**Detecting Hidden Biases in Clinical AI: A Framework for Fair Stroke Outcome Prediction Using
Clinical Data**

Alfredo Varela Vega, Doctor of Philosophy in Biochemistry

Earlel Thiyagaratnam, Doctor of Philosophy in Health Information Science

Keaton Banik, Master of Computer Science

Mohammad Ahsan Siddiqui, Master of Computer Science

Willow Han, Doctor of Philosophy in Neuroscience

Western University

Department of Data Science

April 9, 2025

Abstract

Machine learning (ML) and artificial intelligence (AI) tools in healthcare can unintentionally worsen health disparities. For example, systems trained on data from urban hospitals often perform poorly for rural or low-income patients, leading to delayed diagnoses or treatments. Additionally, outdated medical data used to train ML models may not reflect modern practices, causing recommendations that misalign with current standards. These issues risk deepening inequities, as marginalized groups already face higher rates of conditions like stroke and limited access to care.

Our project addresses these challenges by creating a framework to ensure AI tools remain fair and effective for all populations. We integrate continuous checks for bias at every stage from initial data collection to real-world deployment. This includes considering diverse patient scenarios to test the ML performance in underrepresented conditions. Also, we propose to bring together clinicians, data scientists, policymakers, and patient advocates to define fairness priorities and translate them into actionable metrics.

Our approach prevents AI from perpetuating systemic biases, ensuring tools work equitably across diverse communities. It establishes adaptable standards for AI development, requiring ongoing evaluation similar to drug safety monitoring to keep pace with evolving medical practices. By bridging technical innovation with ethical healthcare delivery, this work aims to build trust in AI, reduce disparities in life-saving treatments, and set a global benchmark for responsible AI in medicine. Ultimately, it ensures AI advancements benefit everyone, not just the most represented groups.

Introduction

Artificial intelligence (AI) and machine learning (ML) are driving transformative advancements in healthcare, offering innovative solutions to improve clinical decision-making and health outcomes (Hasanzadeh et al., 2025; Ueda et al., 2023). These technologies have demonstrated significant potential in specialized fields such as radiology, neurology, and cardiology, where they have shown improvements in diagnostic accuracy and treatment protocols (Hasanzadeh et al., 2025; Ueda et al., 2023). Recent developments in generative AI and large-language models have enabled practical applications of synthetic patient data generation and text-based clinical reasoning support (Kanjee et al., 2023; Kothari, 2023; Lan et al., 2020; Arora & Arora, 2022). By mid-2024, the U.S. Food and Drug Administration (FDA) had approved over 850 AI-enabled medical devices, with nearly 200 added in the preceding year alone, reflecting their rapid integration into clinical practice (Hasanzadeh et al., 2025). However, as these tools grow in complexity and clinical impact, their diminishing explainability raises critical concerns among clinicians, ML engineers, and policymakers regarding algorithmic fairness and equitable benefit distribution (Hasanzadeh et al., 2025; Ueda et al., 2023; Pinsky et al., 2024).

A central challenge lies in the inherent biases of training data, which often reflect systemic disparities in traditional healthcare systems. Existing inequities in resource allocation, access to care, and health outcomes are inadvertently encoded into AI models trained on such data, leading to biased predictions that exacerbate disparities (Hasanzadeh et al., 2025; Ueda et al., 2023). For example, AI-driven diagnostic improvements may disproportionately benefit populations well-represented in training datasets, while underrepresented groups, such as rural or low-income communities, experience diminished returns (Hasanzadeh et al., 2025; Ueda et al., 2023; Pinsky et al., 2024). This issue is particularly evident in stroke care, where predictive models trained on urban hospital datasets like MIMIC-III (Johnson et al., 2016) and MIMIC-IV (Johnson et al., 2023) exhibit performance gaps when applied to rural populations. Obermeyer et al. (2019) demonstrated that such models may fail to account for disparities in diagnostic tool availability or treatment access, resulting in inaccurate predictions for underserved patients. These technical limitations carry profound societal implications, as rural and low-income communities already face higher stroke mortality rates (Saver, 2006). Biased algorithms risk worsening outcomes by delaying time-sensitive interventions like thrombectomy, where timely administration is critical to preventing irreversible neurological damage (Saver, 2006).

The evolution of clinical practices further complicates model reliability. Shifts in stroke treatment protocols over time, such as the adoption of advanced thrombectomy techniques, are not always captured in historical training data. Models trained on outdated datasets may misalign with current guidelines, leading to suboptimal recommendations for modern patient cohorts. This emphasizes the need for continuous bias monitoring and adaptive frameworks to ensure AI tools remain relevant and equitable across diverse and evolving healthcare contexts (Johnson et al., 2016; Johnson et al., 2023; Obermeyer et al., 2019; Saver, 2006).

The Medical Information Mart for Intensive Care (MIMIC-III) (2001–2012) and MIMIC-IV (2008–2019) datasets capture evolving clinical practices, such as the adoption of advanced thrombectomy therapies (Johnson et al., 2016; Johnson et al., 2023). However, models trained on historical data may misalign with contemporary patient cohorts, particularly when clinical guidelines have shifted. For example, algorithms based on outdated treatment protocols may exclude patients eligible for newer interventions, resulting in suboptimal care pathways, increased morbidity, and elevated healthcare costs (Obermeyer et al., 2019; Saver, 2006). Regulatory agencies such as the FDA now mandate fairness assessments for AI tools to preempt harm, emphasizing proactive bias detection rather than post-deployment corrections (Silva et al., 2022).

Current AI/ML development processes often lack interdisciplinary collaboration, a gap that perpetuates clinically irrelevant or biased outcomes. Regulatory bodies, including the WHO and FDA, advocate for robust frameworks to identify, quantify, and mitigate algorithmic biases, alongside legislation to enforce equitable AI deployment (Silva et al., 2022; Gichoya et al., 2023). To address these challenges, this project proposes a systematic methodology to detect and address multivariate biases in clinically deployed AI/ML models. The framework integrates continuous bias monitoring across the model's lifecycle, from data curation and training to post-market surveillance, ensuring fairness metrics adapt to evolving healthcare contexts. Additionally, it emphasizes the evaluation of existing tools for quantifiable bias, providing actionable strategies to mitigate disparities in decision-making processes.

Central to this initiative is the formation of interdisciplinary teams comprising clinicians, data scientists, policymakers, social scientists, and patient advocates. By embedding diverse perspectives into the bias mitigation framework, the project aims to ensure that AI/ML tools deliver generalizable clinical benefits across all populations. This collaborative approach bridges gaps between technical innovation and real-world healthcare needs, fostering trust in AI systems while aligning their development with ethical and equitable care standards (Silva et al., 2022; Gichoya et al., 2023).

Methods

Our study aimed to rigorously evaluate a high-performance ML model for disease classification to determine whether it exhibited biases against specific subgroups within its training data and to quantify the extent of such biases. Supervised learning, widely adopted in healthcare due to its precision in diagnostic and prognostic tasks, served as the foundation for this investigation. We selected a model developed through a peer-reviewed study published in a leading healthcare AI journal, ensuring methodological rigor and clinical relevance (Habeheh & Gohel, 2021).

The research involved four sequential phases. First, we conducted a comprehensive review of bias types prevalent in healthcare ML, including historical, representation, algorithmic, evaluation, aggregation (Mehrabi et al., 2021), and measurement biases (Scarone et al., 2024). Next, we established criteria for model selection: the model needed to address a critical healthcare classification challenge, originate from a reputable study with transparent methodology, and utilize training data enriched with demographic, socioeconomic, and temporal variables to facilitate subgroup analysis and bias detection. Third, we replicated the model using the original study's protocols to ensure fidelity to its design and performance benchmarks. Finally, we systematically tested the replicated model for identified biases, generating a detailed report that highlighted affected subgroups and the magnitude of bias impact on their outcomes.

Our study selected a ML model proposed by Zhang et al. (2022) in *Frontiers in Aging Neuroscience*, which aimed to predict postoperative stroke risk in elderly patients. The model was trained using the MIMIC dataset, a publicly available repository of deidentified health records from patients admitted to emergency departments or intensive care units at Beth Israel Deaconess Medical Center in Boston, Massachusetts. MIMIC-IV, the latest iteration, includes data from over 65,000 ICU admissions and 200,000 emergency department visits spanning 2008 to 2022, providing a robust temporal scope to assess historical biases and demographic subgroup disparities (Johnson et al., 2023). The dataset's inclusion of sociodemographic variables, such as age, race, and socioeconomic indicators, enabled targeted analyses to evaluate whether the model exhibited inequitable performance across patient subgroups.

However, working with the MIMIC dataset presented significant logistical and technical challenges. Access required a multi-step approval process involving institutional endorsements, completion of the Collaborative Institutional Training Initiative (CITI) program, and detailed project proposals, which delayed data acquisition. Additionally, the dataset's size, over 250,000 patient records with high-dimensional clinical variables, demanded substantial computational resources for processing and analysis, exceeding the capacity of standard research infrastructure. A critical barrier arose from the original study's lack of transparency: no replication code, GitHub repository, or explicit methodological details were provided to reconstruct the training and testing datasets. This omission, combined with the dataset's complexity, rendered faithful replication impractical without specialized computational tools. Consequently, we pivoted to a smaller, more manageable dataset to ensure methodological feasibility while preserving the study's focus on bias detection. These challenges underscore the importance of open science practices in healthcare AI research, including code sharing and detailed methodological documentation.

Our study transitioned to the National Health and Nutrition Examination Survey (NHANES) Continuous dataset, curated by the U.S. Centers for Disease Control and Prevention (CDC), to evaluate the ML model's generalizability and bias patterns. Spanning from 1999 to 2018, this dataset encompasses diverse variables, including demographic, socioeconomic, dietary, and health-related metrics. For our proof-of-concept analysis, we prioritized demographic and socioeconomic factors, selecting seven variables: U.S. citizenship, education level, household size, marital status, poverty income ratio (PIR), gender, and age. Following protocols from the original study, we excluded individuals younger than 55 years and stratified the data into stroke ($n = 987$) and non-stroke ($n = 13,557$) cohorts to align with the target clinical population.

To enhance the model's capacity to detect subgroup-specific biases, we introduced a feature combination stage. This process generated all possible permutations of the seven demographic variables, enabling granular analysis of interactions between sociodemographic factors. Subsequent preprocessing mirrored the original methodology: iterative singular value decomposition (SVD) addressed missing data, Synthetic Minority Over-sampling Technique for Nominal and Continuous (SMOTENC) balanced class distributions (with SMOTE or SMOTEN applied where SMOTENC was unsuitable), and normalization standardized inputs prior to model training. We replicated eight ML algorithms from the reference study, decision tree (DT), K-nearest neighbors (KNN), logistic regression (LR), multi-layer perceptron (MLP), random forest (RF), support vector machine (SVM), and XGBoost (XGB), to assess consistency in performance and bias patterns across methodologies. Figure 1 illustrates the integrated pipeline, emphasizing the systematic integration of feature engineering and algorithmic workflows.

Proof-of-Concepts Results

Our proof-of-concept analysis sought to evaluate potential biases in machine learning models through an intersectional lens, focusing on how combinations of demographic and socioeconomic factors influence model performance. Results revealed that models trained exclusively on age consistently achieved high predictive accuracy (Figure 2), suggesting age may dominate decision-making processes at the expense of other variables. While age itself is not inherently biased, this disproportionate reliance

raises concerns that models might undervalue or overlook critical sociodemographic factors, warranting deeper investigation into whether such prioritization perpetuates inequities in clinical outcomes.

Further analysis demonstrated the complexity of multi-factor interactions. Combinations of U.S. citizenship, household size, and gender exhibited poor predictive performance, but integrating poverty income ratio (PIR) and age significantly improved accuracy. This contrast underscores the necessity of examining biases across intersecting variables rather than isolated factors. For instance, models prioritizing age and economic status may inadvertently marginalize subgroups defined by citizenship or family structure, amplifying disparities in healthcare access or resource allocation. These findings emphasize the need for granular, intersectional bias assessments to ensure models do not reinforce systemic inequities through oversimplified variable hierarchies.

Our proposed bias-improvement framework pinpoints feature subgroups that carry a high risk of bias and suggests an alternative output for those subgroups. This secondary output, presented in a user-friendly format, must distinguish between features genuinely essential for decision-making and those that may be biased. While data scientists and machine learning engineers will ultimately refine the algorithm, an initial assessment could be to treat feature subgroups whose removal significantly alters the model's predictions as key to its decision-making. If a particular subgroup yields consistently different performance or results when examined in isolation, it can be flagged as potentially biased. For example, a Parkinson's disease diagnosis model showing reduced accuracy for individuals over 65 might highlight that age subgroup as having a potential bias risk. This workflow is illustrated in Figure 3.

Multidisciplinary Engagement

Our project proposes a generalizable framework for detecting bias in healthcare AI across various stages of development (Figure 3). To effectively identify and mitigate biases, interdisciplinary collaboration involving clinicians, data scientists, policymakers, medical historians, social scientists, medical sociologists, and patients from underrepresented populations is essential. Each stakeholder group provides unique insights that ensure the AI solutions developed are inclusive, effective, and equitable.

Clinicians are crucial stakeholders, providing expertise on disease presentations and ensuring AI models account for nuanced clinical conditions. They guide strategies such as synthetic data generation to enhance model training and will assist with external validation by identifying diverse patient subgroups for model evaluation (Figure 3; Cha et al., 2021). For instance, a model trained on adult data could be tested on pediatric cases of the same disease, ensuring its applicability across different patient populations (Wolff et al., 2019). Medical historians address the concept of shift bias arising from evolving medical terminology and disease classification. Their involvement ensures training datasets remain current, accurate, and relevant, preventing outdated medical concepts from compromising model performance. Social scientists and medical sociologists identify and mitigate systemic biases, especially those impacting underrepresented groups. They ensure social determinants of health (SDOH) such as race, socioeconomic status, and healthcare access, are integrated into model development, enhancing equity and fairness (Hasanzadeh et al., 2025). Incorporating patients from underrepresented populations ensures AI solutions reflect diverse real-world experiences and effectively address healthcare disparities. Their direct involvement throughout the development process strengthens the relevance and inclusivity of AI models. Data scientists and engineers collaborate closely with all stakeholder groups, implementing their insights into technically robust and adaptable AI models capable of addressing biases effectively in clinical practice. Finally, policymakers are essential for the long-term success and ethical implementation of AI models. They should regulate AI tools similarly to medical devices, requiring periodic evaluations and ongoing scrutiny through structured clinical trials and surveillance mechanisms. Such oversight ensures continuous improvement, ethical compliance, and sustained clinical effectiveness post-deployment.

A robust plan for collaboration and stakeholder engagement is essential for this project's success. Stakeholders will actively participate throughout each phase, providing continuous feedback and guidance. Initially, clinicians, patients, and social scientists will collaborate to clearly define clinical

needs, identify potential biases, and determine relevant social determinants of health (SDOH). This early engagement ensures the AI model is grounded in real-world healthcare challenges and addresses diverse factors influencing health outcomes.

During the development phase, data scientists and engineers will closely partner with clinical experts and medical historians to ensure the AI model incorporates accurate, current medical knowledge and remains robust across varied patient populations. Continuous expert feedback will facilitate timely model improvements. In the validation and evaluation phase, policymakers, clinicians, and patient representatives will jointly establish performance metrics and rigorously test the model using diverse clinical scenarios, ensuring accuracy, fairness, and compliance with ethical standards and regulatory requirements. In post-deployment, policymakers will lead ongoing evaluations through surveillance mechanisms, informed by stakeholder feedback, to maintain the model's relevance, effectiveness, and fairness.

Beyond these structured phases, maintaining a clear, shared vision will further strengthen team dynamics. Assigning roles based on expertise, addressing knowledge gaps, and embracing new contributors will enhance team capabilities. Establishing dedicated collaboration spaces, securing collective funding, setting realistic timelines, adapting to challenges, and regularly evaluating progress will foster a culture of continuous improvement and accountability (Sterbenz et al., 2019). This multidisciplinary collaboration will result in an AI solution that is technically robust, socially responsible, and inclusive, effectively mitigating biases and serving diverse patient populations.

Novel Contributions

This project introduces a novel methodology for detecting biases in healthcare ML models by embedding bias assessment across the entire model lifecycle, from data selection and training to deployment and post-market monitoring, rather than relying solely on retrospective adjustments. Current tools, such as IBM Fairness 360, predominantly address biases after models are deployed, overlooking how biases evolve due to temporal shifts in clinical practices or geographic disparities in care access (Bellamy et al., 2019). For instance, Saver (2006) documented significant advancements in stroke treatment, such as thrombectomy, which have altered patient outcomes over time. Similarly, Obermeyer et al. (2019) demonstrated that urban-rural disparities in healthcare resources lead to inequitable model performance. While recent studies have examined demographic biases in stroke prediction models, few address how temporal changes in treatment protocols or geographic care disparities propagate biases. Our framework uniquely integrates these considerations during early development stages, enabling proactive mitigation rather than reactive fixes.

A key innovation lies in considering missing features, such as underrepresented clinical scenarios for delayed neuroimaging access in rural hospitals, which are absent from datasets like MIMIC-IV. Building on methods validated by Juwara et al. (2024), this approach uncovers biases that conventional validation processes might miss. Additionally, the project incorporates an interdisciplinary collaboration model such that neurologists define clinical fairness criteria (e.g., equitable treatment timelines), while data scientists translate these priorities into quantifiable bias metrics. This synergy ensures that bias detection aligns with real-world clinical challenges, addressing a critical gap in prior research that treated fairness as a purely technical task divorced from healthcare inequities.

The framework's adaptability further distinguishes it from existing solutions. Unlike rigid tools focused on narrow technical fixes, our approach tailors bias detection to diverse healthcare contexts. For example, in resource-limited settings, it prioritizes biases linked to insurance status or socioeconomic factors, whereas academic settings might focus on temporal discrepancies between MIMIC-III and MIMIC-IV data. By incorporating stakeholder feedback, clinicians, policymakers, and patients, the framework ensures relevance across environments, from community hospitals to research institutions. This flexibility advocates for redefining model success beyond aggregate accuracy to emphasize equitable performance across all subgroups, a paradigm shift critical for ethical AI deployment in healthcare.

References

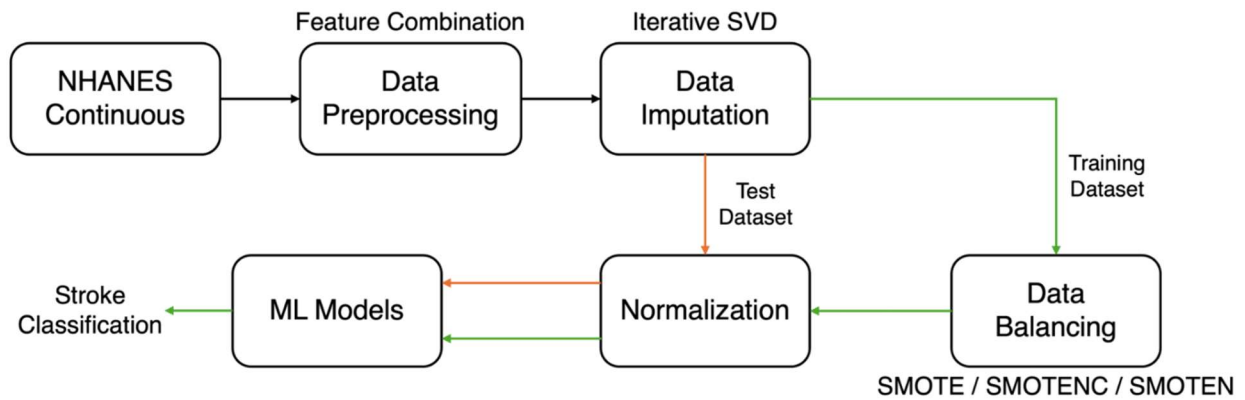
- Arora, A., & Arora, A. (2022). Generative adversarial networks and synthetic patient data: current challenges and future perspectives. *Future Healthcare Journal*, 9(2), 190–193. <https://doi.org/10.7861/fhj.2022-0013>
- Bellamy, R. K. E., Dey, K., Hind, M., Hoffman, S. C., Houde, S., Kannan, K., Lohia, P., Martino, J., Mehta, S., Mojsilovic, A., Nagar, S., Ramamurthy, K. N., Richards, J., Saha, D., Sattigeri, P., Singh, M., Varshney, K. R., & Zhang, Y. (2019). AI Fairness 360: An extensible toolkit for detecting and mitigating algorithmic bias. *IBM Journal of Research and Development*, 63(4/5), 4:1–4:15. <https://doi.org/10.1147/JRD.2019.2942287>
- Cha, D., Pae, C., Lee, S. A., Na, G., Hur, Y. K., Lee, H. Y., Cho, A. R., Cho, Y. J., Han, S. G., Kim, S. H., Choi, J. Y., & Park, H.-J. (2021). Differential Biases and Variabilities of Deep Learning-Based Artificial Intelligence and Human Experts in Clinical Diagnosis: Retrospective Cohort and Survey Study. *JMIR Medical Informatics*, 9(12), 1–18. <https://doi.org/10.2196/33049>
- Da Silva, M., Flood, C. M., Goldenberg, A., & Singh, D. (2022). Regulating the Safety of Health-Related Artificial Intelligence. *Healthcare Policy*, 17(4), 63–77. <https://doi.org/10.12927/hcpol.2022.26824>
- Gichoya, J. W., Thomas, K., Celi, L. A., Safdar, N., Banerjee, I., Banja, J. D., Seyyed-Kalantari, L., Trivedi, H., & Purkayastha, S. (2023). AI pitfalls and what not to do: mitigating bias in AI. *British Journal of Radiology*, 96(1150), 1–8. <https://doi.org/10.1259/bjr.20230023>
- Habehh, H., & Gohel, S. (2021). Machine Learning in Healthcare. *Current Genomics*, 22(4), 291–300. <https://doi.org/10.2174/1389202922666210705124359>
- Hasanzadeh, F., Josephson, C. B., Waters, G., Adedinsowo, D., Azizi, Z., & White, J. A. (2025). Bias recognition and mitigation strategies in artificial intelligence healthcare applications. *NPJ Digital Medicine*, 8(1), 1–13. <https://doi.org/10.1038/s41746-025-01503-7>
- Hong, C., Pencina, M. J., Wojdyla, D. M., Hall, J. L., Judd, S. E., Cary, M., Engelhard, M. M., Berchuck, S., Xian, Y., D'Agostino, S., Howard, G., Kissela, B., & Henao, R. (2023). Predictive Accuracy of Stroke Risk Prediction Models Across Black and White Race, Sex, and Age Groups. *JAMA : The Journal of the American Medical Association*, 329(4), 306–317. <https://doi.org/10.1001/jama.2022.24683>
- Johnson, A. E. W., Bulgarelli, L., Shen, L., Gayles, A., Shammout, A., Horng, S., Pollard, T. J., Hao, S., Moody, B., Gow, B., Lehman, L. H., Celi, L. A., & Mark, R. G. (2023). MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10(1), 1–9. <https://doi.org/10.1038/s41597-022-01899-x>
- Johnson, A. E. W., Pollard, T. J., Shen, L., Lehman, L. H., Feng, M., Ghassemi, M., Moody, B., Szolovits, P., Anthony Celi, L., & Mark, R. G. (2016). MIMIC-III, a freely accessible critical care database. *Scientific Data*, 3(1), 1–9. <https://doi.org/10.1038/sdata.2016.35>
- Juwara, L., El-Hussuna, A., & El Emam, K. (2024). An evaluation of synthetic data augmentation for mitigating covariate bias in health data. *Patterns (New York, N.Y.)*, 5(4), 1–19. <https://doi.org/10.1016/j.patter.2024.100946>

- Kanjee, Z., Crowe, B., & Rodman, A. (2023). Accuracy of a Generative Artificial Intelligence Model in a Complex Diagnostic Challenge. *JAMA : The Journal of the American Medical Association*, 330(1), 78–80. <https://doi.org/10.1001/jama.2023.8288>
- Kothari, A. N. (2023). ChatGPT, Large Language Models, and Generative AI as Future Augments of Surgical Cancer Care. *Annals of Surgical Oncology*, 30(6), 3174–3176. <https://doi.org/10.1245/s10434-023-13442-2>
- Lan, L., You, L., Zhang, Z., Fan, Z., Zhao, W., Zeng, N., Chen, Y., & Zhou, X. (2020). Generative Adversarial Networks and Its Applications in Biomedical Informatics. *Frontiers in Public Health*, 8, 1–14. <https://doi.org/10.3389/fpubh.2020.00164>
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2022). A Survey on Bias and Fairness in Machine Learning. *ACM Computing Surveys*, 54(6), 1–35. <https://doi.org/10.1145/3457607>
- Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science (American Association for the Advancement of Science)*, 366(6464), 447–453. <https://doi.org/10.1126/science.aax2342>
- Pinsky, M. R., Bedoya, A., Bihorac, A., Celi, L., Churpek, M., Economou-Zavlanos, N. J., Elbers, P., Saria, S., Liu, V., Lyons, P. G., Shickel, B., Toral, P., Tscholl, D., & Clermont, G. (2024). Use of artificial intelligence in critical care: opportunities and obstacles. *Critical Care (London, England)*, 28(1), 1–12. <https://doi.org/10.1186/s13054-024-04860-z>
- Saver, J. L. (2006). Time is brain-quantified. *Stroke (1970)*, 37(1), 263–266. <https://doi.org/10.1161/01.STR.0000196957.55928.ab>
- Scarone, B., Viola, A., Miller, R. J., & Baeza-Yates, R. (2024). A Principled Approach for a New Bias Measure. *ArXiv.Org*.
- Sterbenz, J. M., Nasser, J. S., & Chung, K. C. (2019). Organizing a Multidisciplinary Research Team: Strategies, Execution, and Outcomes. *Plastic and Reconstructive Surgery (1963)*, 143(3), 951–961. <https://doi.org/10.1097/PRS.00000000000005326>
- Ueda, D., Kakinuma, T., Fujita, S., Kamagata, K., Fushimi, Y., Ito, R., Matsui, Y., Nozaki, T., Nakaura, T., Fujima, N., Tatsugami, F., Yanagawa, M., Hirata, K., Yamada, A., Tsuboyama, T., Kawamura, M., Fujioka, T., & Naganawa, S. (2024). Fairness of artificial intelligence in healthcare: review and recommendations. *Japanese Journal of Radiology*, 42(1), 3–15. <https://doi.org/10.1007/s11604-023-01474-3>
- Wolff, R. F., Moons, K. G. M., Riley, R. D., Whiting, P. F., Westwood, M., Collins, G. S., Reitsma, J. B., Kleijnen, J., & Mallett, S. (2019). PROBAST: A Tool to Assess the Risk of Bias and Applicability of Prediction Model Studies. *Annals of Internal Medicine*, 170(1), 51–58. <https://doi.org/10.7326/M18-1376>
- Zhang, X., Fei, N., Zhang, X., Wang, Q., & Fang, Z. (2022). Machine Learning Prediction Models for Postoperative Stroke in Elderly Patients: Analyses of the MIMIC Database. *Frontiers in Aging Neuroscience*, 14, 1–9. <https://doi.org/10.3389/fnagi.2022.897611>

Appendix

Figure 1

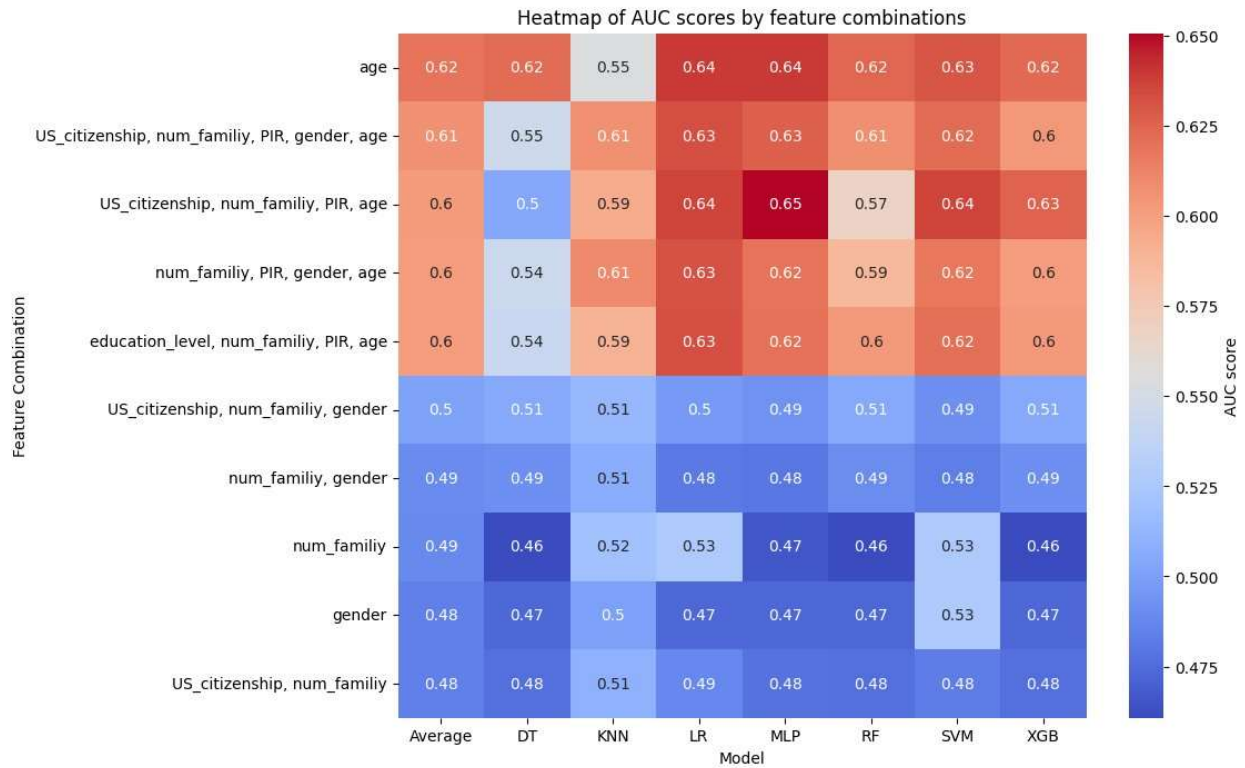
Proposed Stroke Prediction Model



Note. A schematic illustration of the stroke prediction model by Zhang et al. (2022).

Figure 2

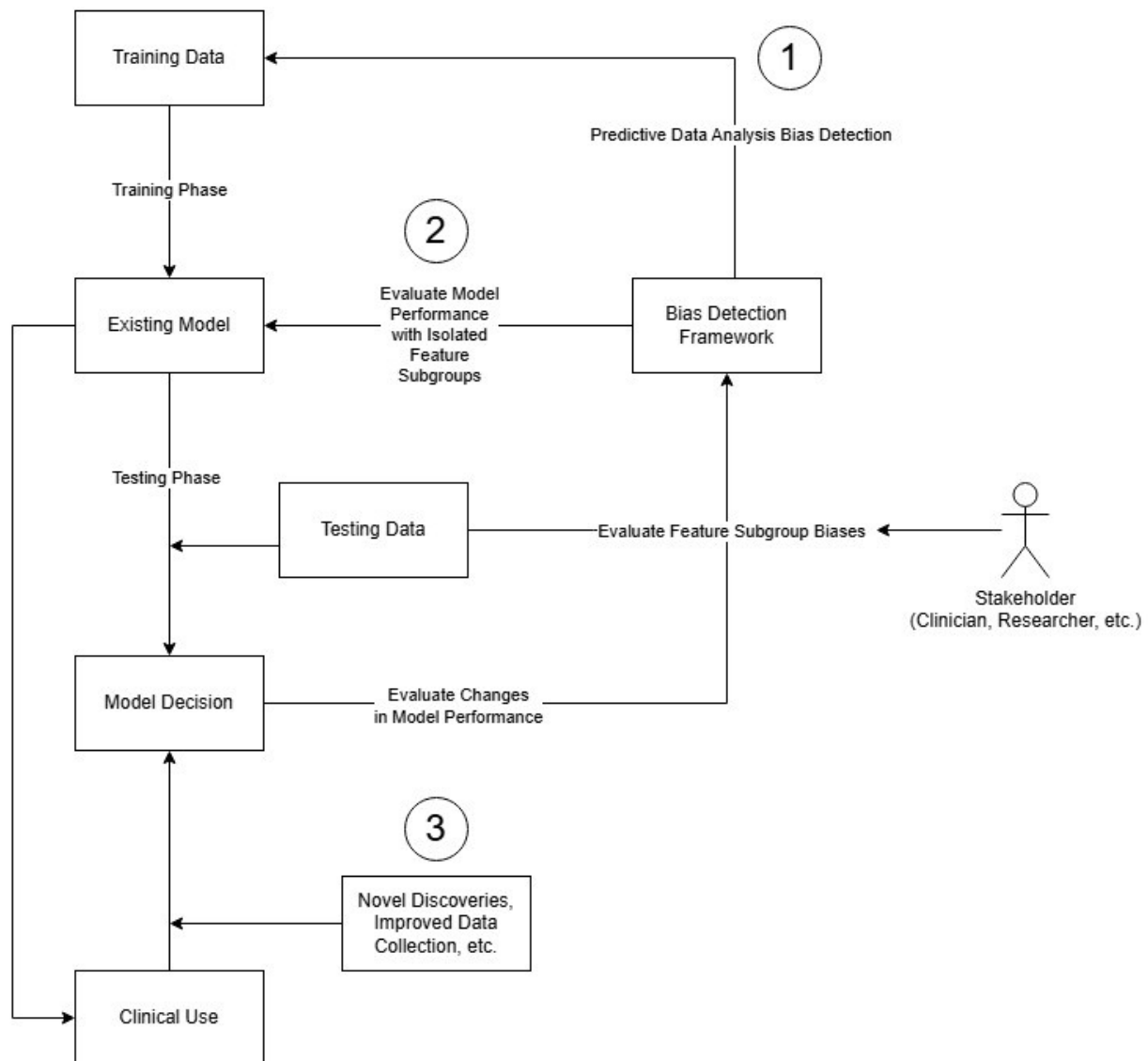
Feature Heatmap of AUC Scores



Note. Heatmap showing the AUC scores for the top five and bottom five combinations of demographic and socioeconomic features. The scores are colour-coded using a red-white-blue gradient: red represents high AUC scores, blue indicates low AUC scores, and white corresponds to mid-range AUC scores.

Figure 3

Bias Detection and Mitigation Framework Across the Model Lifecycle



Note. This diagram outlines three key stages in the lifecycle of a ML/AI model where bias can be identified and addressed (additional stages may also apply). 1) Training Phase, Predictive Data Analysis: Initial biases, such as class imbalances or systemic data collection issues, are detected prior to model development. 2) Testing Phase, Bias Evaluation and Stakeholder Feedback: Multivariable and nonlinear biases are uncovered by iteratively evaluating model performance across isolated feature subgroups using a bias detection framework. This process incorporates stakeholder input (clinicians, researchers, etc.) and informs both model parameters and interpretation of results. 3) Clinical Use, Ongoing Monitoring and Adaptation: During real-world deployment, evolving factors such as new data patterns, treatments, or population shifts are assessed for their impact on model performance. Similar methods from step 2 are used to evaluate and adapt the model accordingly.